

Hector Kohler

hector.kohler@polytechnique.edu
kohlerhector.github.io/homepage/
github

Research Experience

- 2026-2027 Postdoc researcher on reinforcement learning and graphs, [LIX](#), École Polytechnique, (Advisors: [Dr. J. Lutzeyer](#) and [Pr. J. Read](#))
- 2025 Visiting researcher designing new RL algorithms and evaluation tools, [RLAI Lab](#), University of Alberta, (Advisor: [Dr. A. White](#))
- 2022-2025 Ph.D. researcher on interpretable reinforcement learning, [Inria School](#), Université de Lille, (Advisors: [Dr. R. Akroure](#) and [Pr. P. Preux](#))
- 2021 Research assistant working on unmanned underwater drone control, [Lab-STICC](#), ENSTA, (Advisors: [Pr. B. Clement](#) and [G. Le Chenadec](#))

Education

- 2022-2025 **Ph.D.**, Interpretable Reinforcement Learning, Université de Lille
- 2020-22 **M.Sc.**, Artificial Intelligence *Excellence Track*, Sorbonne Université
- 2016-20 **B.Sc.**, Computer Science & Mathematics, University of Manchester

Awards & Honors

- 2025 [Université de Lille International Mobility Grant](#) (1650 €)
- 2022 [AI_PhD Lille](#) (Competitive admission to fully funded Ph.D. program)
- 2022 Sorbonne Université Computer Science Excellence diploma (10/400 students)

Publications

 [Google Scholar](#)

Journal Articles

- J1. Mathieu, T., Medeiros Centa, M., Della Vecchia, R., **Kohler, Hector**, Shilova, A., Maillard Maillard, O.-A. & Preux, P. AdaStop: adaptive statistical testing for sound comparisons of Deep RL agents. *Transactions on Machine Learning Research* (2024).
- J2. **Kohler, Hector**, Clement, B., Chaffre, T. & Le Chenadec, G. PID Tuning using Cross-Entropy Deep Learning: a Lyapunov Stability Analysis. *IFAC-PapersOnLine* (2022).

Peer-reviewed Conference Proceedings

- C1. **Kohler, Hector**, Akroure, R. & Preux, P. *Limits of reinforcement learning for decision trees in Markov decision processes* in *Reinforcement Learning Conference* (2026). <https://openreview.net/forum?id=BBeszTCQhT>.

- C2. **Kohler, Hector**, Akrou, R. & Preux, P. *Breiman meets Bellman: Non-Greedy Decision Trees with MDPs* in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (2025).

Posters

- P1. **Kohler, Hector**, Delfosse, Q., Radji, W., Akrou, R. & Preux, P. *Evaluating Interpretable Reinforcement Learning by Distilling Policies into Programs* Workshop on Programmatic Reinforcement Learning (Edmonton, Canada). 2025.
- P2. **Kohler, Hector**, Delfosse, Q., Akrou, R., Kersting, K. & Preux, P. *Interpretable and editable programmatic tree policies for reinforcement learning* 17th European Workshop on Reinforcement Learning (Toulouse, France). 2024.

Tools & Software

[Core contributor to rlberry](#): PyTorch implementation of deep reinforcement learning agents. Compatibility with `stable-baselines3`. Advanced training analysis with robust stats. Usage of git for development and continuous integration. Organized hackatons.

[Maintainer of a set of fully-transparent control policies for games and robots](#): inspired by the `stable-baselines3-zoo` and the `gymnasium` API.

[Designed statistical test for the reliable and cheap comparison and benchmarking of RL](#).

[Maintainer of my own scikit-learn estimator](#): I developped DPDT, a decision tree induction algorithm which is SOTA for tabular problems.

Teaching

Ecole Normale Supérieure de Cachan (MVA)

2024 Assistant Instructor, Reinforcement Learning (6 hours)

Centrale Lille

2024 Assistant Instructor, Sequential Decision Making (6 hours)

Université de Lille

2023-2024 Assistant Instructor, Algorithms and Programming (60 hours)

2022 Assistant Instructor, Reinforcement Learning (24 hours)

Academic Advising

Graduate

2023	Veronika Shilova, Ecole Polytechnique, Entropy-regularized RL
2023	Quentin Dinel, Université de Lille, Image Segmentation
2023	Mathieu Medeng Essia, Centrale Lille, Imitation Learning
2023	Antonio Al-Makdissi, Centrale Lille, Object-centric Atari Learning
2023	Gabriele Maggioni, Centrale Lille, Model-Based Reinforcement Learning

Academic Service

Reviewing

2026	RLC, TMLR
2025	ICLR, RLC, AISTATS, ICML
2024	NeurIPS

Talks

2025	Seminar at ENSTA Paris for Sao MAI NGUYEN group.
2025	RL seminar at the University of Alberta in Adam White group.
2025	Orals at RLC and KDD.
2023- 2024	Scool seminars.

Chairing

2026	Program chair in charge of OpenReview for EWRL 2026 .
2024	Organized the InterPol workshop at the first Reinforcement Learning Conference in Amherst (USA).

Responsibilities

2023-2025	Elected representative of Ph.D. students and Postdocs at the CRISAL laboratory (Université de Lille).
2025	Ph.D. representative for the HCERES evaluation of CRISAL.

Last updated: May 17, 2026